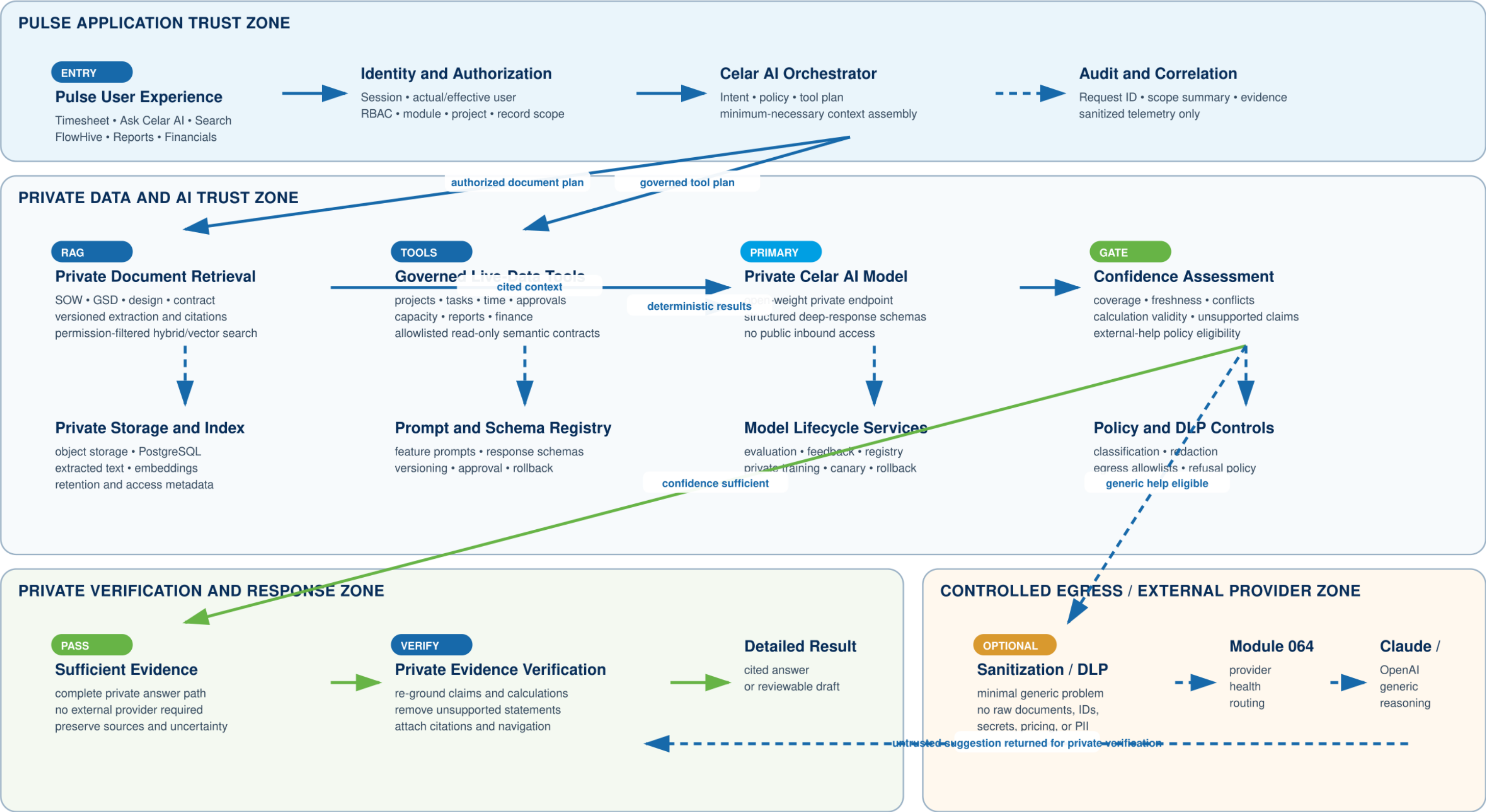


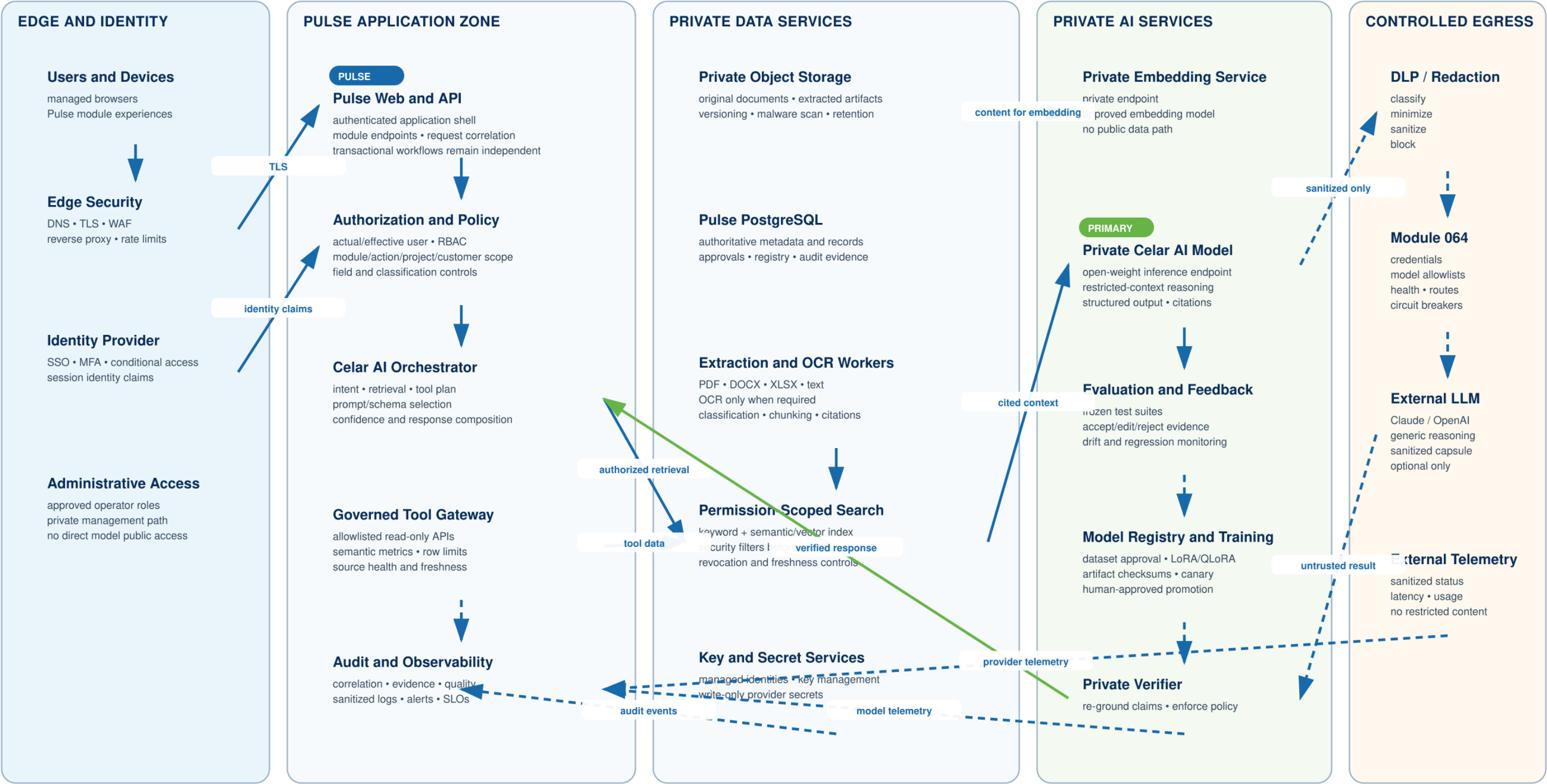
Celar AI Private-First Logical Architecture — Speed of Delivery

US Signal • Pulse • Module 011 • Celeritas to speed of delivery • Version 2.0



Celar AI Deployment and Network Architecture

US Signal • Pulse • Module 011 • Celeritas to speed of delivery • Version 2.0



US Signal Internal — Confidential • Private endpoints and outbound allowlists enforce the trust zones. External providers never receive unrestricted Pulse context.

Celar AI OpenCloud Private Runtime Architecture

US Signal • Pulse • Module 011 • Celeritas to speed of delivery • Version 2.0

CURRENT INTERIM STATE — NO ADDITIONAL AZURE PRIVATE-RUNTIME RESOURCES

Existing Pulse Test

Web/API • PostgreSQL • Azure Files

CONTINUES

Internal Data Intelligence

Permission-scoped tools and deterministic queries

PAUSED

Document Runtime

OCR • malware • embeddings • inference deferred

FlowHive Guard

No SOW plan until citations are ready

PULSE APPLICATION AND DATA

APP

Pulse Web and API

identity • RBAC • orchestration
governed internal-data tools

DATA

PostgreSQL and Documents

authoritative records • citations
SOW/GSD files • versions • audit

Document Pipeline

scan → extract/OCR → chunk
retrieve → approve → cite

FlowHive

SOW-grounded WBS and schedule
fail closed when evidence is unavailable

PRIVATE LINK

Private Route

DNS • TLS
host allowlist
firewall rules

private HTTPS

Service Auth

secrets
least privilege
rotation

No Public

service ingress
or browser path

OPENCLOUD SHARED TEST/UAT RUNTIME VM

ONE PRIVATE LINUX VM • PODMAN / OCI • TEST/UAT

CONTAINER

Ollama

private inference
private embeddings
CPU first • GPU optional

CONTAINER

Tesseract Adapter

OCR for scanned files
native extraction bypass
bounded worker queue

CONTAINER

ClamAV

malware scanning
signature updates
private TCP/API path

authenticated OCR

private scan

Persistent Volumes

Ollama models • ClamAV signatures • OCR work files • service configuration
containers remain replaceable; evidence remains in governed Pulse storage

Runtime Controls

health probes • resource limits • one-document processing concurrency
service identities • audit correlation • backup and rollback runbook

PRODUCTION EVOLUTION

Ollama Scale-Out

move to dedicated
GPU-capable compute
when load requires

CPU Services

Tesseract + ClamAV
may remain together
scale by queue depth

Stable Contract

same private APIs
placement can change

US Signal Internal — Confidential • Shared-VM placement is the approved Test/UAT starting point. Pulse application and PostgreSQL placement remain independent. Enable FlowHive only after live scan, extraction, retrieval, approval, and citation checks pass.